



Mammary gland responses to supplementation of rumen-protected methionine products in dairy cows

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ABSTRACT

Methionine can be a limiting nutrient in lactating dairy cows consuming diets with corn, corn-based, and soybean-based ingredients. Rumen-protected Met (RPM) supplementation improves productivity in high-producing cows, and the supplementation of RPM may improve productivity in high-producing cows fed moderate-concentration CP diets. The objective of this work was to compare an RPM prototype with a commercially available RPM product on milk protein concentration, plasma Met concentration, and mammary gland clearance of Met in dairy cows. In experiment 1, 10 multiparous cows (DIM = 76 ± 17 [arithmetic mean $\pm$ SD]) were used in a duplicated 5×5 Latin square with 18 d periods. A commercial product and an RPM prototype were used to formulate the treatments as follows: (1) control (CTL1; 0 g/d of RPM supplemented), (2) 22 g/d of Met supplemented as RPM using Smartamine M (SM, Adisseo, Alpharetta, GA), (3) 22 g/d of Met supplemented using RPM prototype 1 (PT1.0), (4) 27 g/d of Met supplemented using RPM prototype 1 (PT1.25), and (5) 33 g/d of Met supplemented using RPM prototype 1 (PT1.5). In experiment 2, 20 multiparous cows (DIM = 72 ± 43 , arithmetic mean $\pm$ SD) were used in a quadruplicate 5×5 Latin square with 14 d periods. Cows were randomly assigned to the following treatments: (1) control (CTL2; 0 g of RPM supplemented), (2) 6 g/d metabolizable Met supplemented using Smartamine M (SM6), (3) 12 g/d metabolizable Met supplemented using Smartamine M (SM12), (4) 6 g/d metabolizable Met supplemented using RPM prototype 2 (PT6), and (5) 12 g/d of metabolizable Met supplemented using RPM prototype 2 (PT12). The CTL1 and CTL2 diets were predicted to be deficient in metabolizable Met. The CTL1 and CTL2 diets contained 16.0% and 16.7%

CP, 2.03% and 2.02% of metabolizable Met, and 6.74% and 6.84% of metabolizable Lys. In experiments 1 and 2, feed ingredients, milk, and blood (coccygeal vessels) samples were collected and analyzed. In experiment 2, abdominal mammary vein blood samples were collected to determine mammary gland clearance. Data were analyzed using mixed models with animal as the experimental unit, treatment and time as fixed effects, and animal as a random effect. In experiment 1, and compared with CTL1, SM increased milk protein concentration. Compared with CTL1, treatment SM increased plasma Met concentrations. Compared with SM, the PT1.0, PT1.25, and PT1.5 treatments had lower plasma Met concentrations. Treatments PT1.5 and SM supported similar milk protein concentration response. In experiment 2, SM12 increased milk protein concentration compared with CTL2, and the prototype treatments sustained milk protein content similar to SM6 and SM12 treatments. Compared with CTL2, supplementation with SM12 increased plasma Met concentrations, and SM12 had greater concentrations than PT12 in the abdominal mammary vein. Compared with CTL2, SM12 decreased Met mammary clearance rate. Supplementation with RPM products may have increased protein synthesis in the udder of lactating cows. In conclusion, although supplementation with RPM products may have increased protein synthesis in the udder of lactating cows, it indeed increased plasma Met concentrations, confirming the ability of RPM products to increase plasma Met. Supplementation with RPM reduced mammary gland clearance of Met, suggesting the use of additional Met supply in other tissues.

Key words: dairy cow, mammary gland, milk protein, rumen-protected methionine

INTRODUCTION

Corn, corn-based byproducts, and soybean meal are common ingredients of dairy cattle diets in North America. These ingredients provide relatively low concentrations of Met to meet cows' requirements to produce milk and milk protein. Various nutritional strategies can

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

be employed to meet metabolizable Met requirements, including providing high dietary CP content, mixing animal-based byproducts in the ration, or dietary supplementation with rumen-protected Met (RPM; Wang et al., 2010). The first 2 strategies are less sustainable and efficient because an excess of dietary CP leads to poor N-use efficiency as well as decreases in Met capture and utilization in the udder and milk protein (Raggio et al., 2004). In high-producing dairy cows fed corn- and soybean-based diets, it is more efficient to provide ruminally undegraded Met as RPM and a single EAA supplementation approach rather than using RUP that provides metabolizable EAA and NEAA in addition to Met (Broderick et al., 2009).

Rumen-protected Met products are developed to limit Met degradation by microorganisms in the rumen. Supplementing the diet with Met protected from rumen degradation may improve metabolizable Met supply and the synthesis of milk protein and fat in dairy cattle (Leonardi et al., 2003; Broderick et al., 2009; Chen et al., 2011). To this end, a meta-analysis using 64 studies showed that supplemental RPM sources or postruminal infusion of Met to dairy cows in early or mid-lactation increased milk protein synthesis (Zanton et al., 2014).

Determining the effect of methionine from RPM supplementation on productive response is necessary to formulate dairy rations, meet metabolizable Met requirements, and achieve optimal milk production and milk component response. The effectiveness of RPM products on lactation performance varies; however, milk protein response and plasma Met concentration could be used to indicate changes in metabolizable Met supply when supplementing RPM products (Zanton et al., 2014; Whitehouse et al., 2016). Therefore, the study objective was to compare an RPM prototype with a commercially available RPM product on milk protein concentration, plasma Met concentrations, and mammary gland clearance of Met in dairy cows. We hypothesized that supplementing RPM would increase milk protein content and plasma Met concentrations, whereas increasing RPM supply would reduce the clearance of AA in the mammary glands. Two separate experiments tested the hypothesis. Prototypes of RPM used were developed for the studies and were not commercially available.

MATERIALS AND METHODS

Experiment 1

Animals, Experimental Design, and Treatment. The Institutional Animal Care and Use Committee of the University of Tennessee approved experimental procedures (2480–0916). Experiment 1 used 10 multiparous Holstein cows (arithmetic mean $\pm$ SD: DIM = 76 $\pm$ 17, and milk

Table 1. Formulated ingredient inclusion and nutrient composition in basal diet (% of DM, unless stated otherwise)

Item	Experiment 1	Experiment 2
Ingredient		
Corn silage	38.5	40.9
Alfalfa hay	13.6	—
Ryegrass silage	—	10.9
Corn grain, ground, dry	14.8	16.5
Soybean meal, solvent (48% CP)	6.45	10.9
Soybean hulls, ground	6.35	4.30
Protected soybean meal ¹	4.60	4.50
Canola meal, solvent	4.45	—
Citrus pulp, dry	4.85	5.10
Molasses, sugarcane	0.50	0.90
Urea	0.25	0.30
Blood meal, treated ²	0.60	0.55
Fat supplement ³	1.75	0.65
Palmit 80 ⁴	—	0.65
Fermentation aid ⁵	0.05	0.05
Probiotic ⁶	0.20	0.20
Monensin ⁷	0.05	0.10
Salt	0.40	0.35
Limestone, ground	0.90	0.90
Calcium phosphate (di-)	0.20	0.25
Sodium bicarbonate	0.70	0.90
Potassium carbonate	0.20	0.30
Potassium magnesium sulfate	0.15	0.20
Trace mineral and vitamin mix ^{8,9}	0.60	0.60
Lysine supplement ¹⁰	<0.01	<0.01
Nutrient composition		
DM, ¹¹ %	47.4	42.0
CP	16.7	16.0
NDF	27.7	29.2
ADF	20.1	20.0
NFC	43.8	41.9
Starch	27.5	26.5
Crude fat	5.16	4.92
NEL, Mcal/kg	1.67	1.68
Met, % of MP	2.03	2.02
Lys, % of MP	6.74	6.82
Ca	0.84	0.81
P	0.36	0.37

¹SoyPLUS, Landus, Des Moines, IA.

²ProvAAI AAdvantage, Perdue AgriBusiness (Salisbury, MD).

³Energy Booster 100, Milk Specialties Global (Eden Prairie, MN).

⁴Natu'oil Services Inc. (Port Coquitlam, BC Canada).

⁵Original XPC, Diamond V Mills Inc. (Cedar Rapids, IA).

⁶OmniGen-AF, Phibro (Stamford, CT).

⁷Rumensin, Elanco (Greenfield, IN).

⁸Experiment 1: AgCentral Cooperative (Athens, TN); basal diet contained 2.7×10^5 IU of vitamin A, 6.2×10^4 IU of vitamin D, 1.9×10^3 IU of vitamin E, 2.5×10^3 mg of Mn, 609 mg of Cu, 3.5×10^3 mg of Zn, 29 mg of I, 18 mg of Co, 9.3 mg of Se.

⁹Experiment 2: AgCentral Cooperative (Athens, TN); basal diet contained 4.9×10^5 IU of vitamin A, 0.8×10^3 IU of vitamin D, 1.3×10^2 IU of vitamin E, 4.1×10^2 mg of Mn, 90.2 mg of Cu, 5.5×10^2 mg of Zn, 6.6 mg of I, 3.8 mg of Co, 0.80 mg of Se.

¹⁰AjiPro-L, Ajinomoto Co. Inc. (Chicago, IL) at 30 g/d of product.

¹¹Actual DM of TMR.

yield = 55.3 ± 4.41 kg/d at the beginning of the study) from the main herd of the East Tennessee AgResearch and Education Center, Little River Animal and Environmental Unit. Cows were housed in the main barn with sand-bed-

Table 2. Experiment 1 analyzed nutrients and chemical composition of the feed ingredients used in the basal diet (% of DM)

Item	Ingredient ¹			
	Corn silage	Alfalfa hay	Soybean hulls	Grain mix
DM, % of feed	30.1	83.1	91.0	91.5
NDF	38.8	35.5	72.7	13.0
ADF	24.2	30.2	49.4	8.53
CP	8.60	22.0	11.0	23.9
Starch	33.1	2.60	—	60.8
Crude fat	3.44	2.76	—	6.07
Ca	0.23	1.88	0.68	1.89
P	0.23	0.28	0.12	0.62
Mg	0.20	0.19	0.28	0.64
K	1.12	3.39	1.48	1.52
Na	0.01	0.03	0.02	1.04

¹Grain mix contained all ingredients listed in Table 1 in addition to corn silage, alfalfa hay, and soybean hulls.

ded freestalls. Cows were stratified based on DIM, milk yield, and parity and randomly assigned to a duplicated 5 × 5 Latin square design. The length of each period was 18 d (14 d of adjustment to treatments and 4 d of data collection). Treatments were as follows: (1) control (CTL1; 0 g/d of RPM supplemented), (2) 22 g/d of Met supplemented as RPM using Smartamine M (SM; Adisseo, Alpharetta, GA), (3) 22 g/d of Met supplemented as RPM prototype 1 (PT1.0), (4) 27 g/d of Met supplemented as RPM prototype 1 (PT1.25), and (5) 33 g/d of Met supplemented as RPM prototype 1 (PT1.5). Smartamine M contained 75% D,L-Met (DM basis) with a bioavailability of 80% (based on NRC, 2001 nutritional specifications). Prototype 1 contained 69% of D,L-Met (DM basis) and 31% fat coating with predicted bioavailability of 58% (Balchem Corp., Montvale, NJ). The prediction in bioavailability was based on Met concentration and metabolizable Met in the prototype using a 3-step in vitro procedure (Ross et al., 2013). Regarding supply of metabolizable Met, CTL1 treatment was predicted to have a small deficiency in metabolizable Met. To determine the effect of the prototype in relation to a commercially available product, PT1.0 and PT1.25 treatments were designed to have lower supply of metabolizable Met compared with SM treatment. The PT1.5 treatment was designed to match the supply of metabolizable Met of the SM treatment.

Supplements were top-dressed once daily, immediately after ~0800 h feeding, and mixed into the top third portion of the diet. Before the start of the study, a basal diet was balanced and formulated for a 669-kg lactating cow consuming 27.0 kg/d of DM and producing 54.0 kg/d of milk with 3.55% milk fat and 3.06% milk protein using NDS Professional (R.U.M.&N Nutritional Dynamic System, Reggio Emilia, Italy; an industry-licensed system of Cornell Net Carbohydrate-Protein System v.6.55, Cornell University, Ithaca, NY). The diet had a 52:48 forage-to-concentrate ratio (Table 1) and

was predicted to have 16.7% CP content and a balance of ~16.4 g/d metabolizable Met (2.03% of MP supply). The forage portion included corn silage and alfalfa hay, and the concentrate portion included corn grain, soybean meal, canola meal, protected soybean meal (SoyPLUS; Landus, Des Moines, IA), soybean hulls, and citrus pulp. During a 14-d standardization period immediately before the assignment of treatments, cows were fed the basal diet supplemented with 22 g of Met from Smartamine M. The basal diet contained 30 g of Aji-Pro L generation 2 (Ajinomoto Co. Inc., Chicago, IL) to meet Lys requirements. The diets were fed to individual cows using an electronic feeding system (American Calan Inc., Northwood, NH) targeting 5% to 10% TMR refusal (as-fed basis). Cows were milked at 0900 and 1930 h in a double-8 herringbone parlor, and yields were recorded at each milking. The BW was recorded once for 2 consecutive days at the beginning and end of the standardization and treatment periods following immediately after morning milking, and the arithmetic mean by period was included in the statistical analysis.

Sample Collection and Analyses. Forages and TMR samples were collected 3 times weekly and pooled. Orts samples were collected d 11, 12, 13, and 14 of each period. Soybean hulls and the grain mix were sampled once weekly. Forages, soybean hulls, and grain mixes were dried, ground, and submitted for chemical analysis (Cumberland Valley Analytical Services, Waynesboro, PA; Tables 2 and 3). Forage samples were analyzed for CP, lignin, starch, sugar, and ether extract using near-infrared spectroscopy (Foss 5000, Foss North America, Eden Prairie, MN). Soybean hulls and the grain mix were analyzed for CP (AOAC International, 1999; method 990.03; LECO Co., Saint Joseph, MI), starch (Hall, 2009), and ether extract (AOAC International, 1999; method 2003.05). Each dietary ingredient was analyzed for NDF (Van Soest et al., 1991), ADF (AOAC International, 1999; method 973.18), and minerals (AOAC International, 2000; method 985.01). Individual dietary components were analyzed for CP (AOAC International, 1999; method 994.12) and AA with ion-exchange chromatography (AOAC International, 1999; method 988.15) by Midwest Laboratories (Omaha, NE; Table 4).

Milk samples were collected at each milking on d 11, 12, 13, and 14 of the standardization periods and on d 15, 16, 17, and 18 of the treatment periods. Milk samples were analyzed for fat, protein, lactose, SNF, and MUN by infrared analyses (Fossomatic FC, Foss, Eden Prairie, MN; United DHIA, Radford, VA), and milk SCC was analyzed by flow cytometry (Fossomatic FC, Foss; United DHIA). Energy-corrected milk was calculated from the equation defined by Tyrrell and Reid (1965).

Blood samples (10 mL) were collected from the coccygeal vessels at 1000 h on d 14 of the standardization

Table 3. Experiment 1 AA content (% of DM) of individual feed ingredients included in the basal diet

Item	Ingredient								
	Corn silage	Alfalfa hay	Corn grain	Soybean hull	Canola meal	Blood meal ¹	Soybean meal	Protected soybean meal ²	Citrus pulp
Ala	0.56	0.92	0.55	0.41	1.55	5.64	1.99	1.82	0.26
Arg	0.18	0.82	0.40	0.48	2.15	3.02	3.36	2.95	0.25
Asp	0.49	2.23	0.52	0.83	2.54	6.98	5.16	4.64	0.56
C-C ³	0.10	0.23	0.20	0.62	0.78	0.76	0.62	0.47	0.08
Glu	0.90	1.59	1.33	1.04	6.07	6.42	8.26	7.27	0.43
Gly	0.30	0.85	0.31	0.74	1.85	3.32	1.97	1.81	0.25
His	0.16	0.38	0.24	0.26	1.02	4.50	1.27	1.14	0.12
Ile	0.27	0.73	0.26	0.36	1.46	0.82	1.92	1.91	0.17
Leu	0.62	1.19	0.83	0.60	2.35	8.25	3.42	3.13	0.29
Lys	0.23	0.83	0.26	0.64	1.97	6.03	2.38	2.29	0.18
Met	0.11	0.24	0.16	0.08	0.76	0.85	0.64	0.55	0.06
Phe	0.31	0.83	0.37	0.34	1.47	5.10	2.31	2.12	0.21
Pro	0.45	1.25	0.62	0.50	2.16	3.22	2.34	2.12	0.55
Ser	0.21	0.80	0.37	0.54	1.56	3.85	2.42	2.09	0.22
Thr	0.22	0.62	0.28	0.24	1.05	3.72	1.75	1.55	0.15
Trp	0.03	0.22	0.05	0.11	0.40	1.08	0.54	0.42	0.04
Tyr	0.24	0.63	0.31	0.45	1.10	1.94	1.72	1.55	0.16
Val	0.36	0.91	0.35	0.43	1.79	5.92	2.02	1.97	0.22

¹ProvAAL AAdvantage, Perdue AgriBusiness (Salisbury, MD).²SoyPLUS (Landus, De Moines, IA).³C-C = cystine.

period and d 18 of each treatment period into 140 IU sodium heparin tubes (Thermo Fisher Scientific, Waltham, MA) and placed directly on ice. The concentration of AA in plasma from coccygeal vessels was used to estimate concentrations in arterial plasma in cattle. The concentration of AA in the mammary abdominal vein was used to conduct arterio-venous difference studies in lactating cattle (Hanigan et al., 1998b). For each experiment, plasma was separated by centrifugation at $1,500 \times g$ for 20 min at 4°C and stored at -80°C until analysis. Liquid chromatography coupled with MS was used to determine concentrations of plasma-free AA in single samples run in duplicates using a commercially available kit (EZ: faast, Phenomenex Co., Torrance, CA) as previously described (Kassube et al., 2017; Kaufman et al., 2018). The total AA CV was 21% (for interassay) and 16% (for intra-assay). Methionine, Phe, and Tyr showed 6.4%, 10%, and 15% interassay CV, respectively.

Calculations and Statistical Analyses. The Latin square experimental design used in the experiment was not balanced for residual effects. Results were analyzed using a split-plot statistical model with repeated measures, and the ANOVA was conducted using a Mixed procedure in SAS (version 9.4; SAS Institute Inc., Cary, NC). Repeated measures were conducted on days within the treatment-period-cow structure, with day and day-by-treatment interactions tested. The experimental unit was the individual cow. The data analysis included DMI, nutrient intake, and milk yield on d 13, 14, 15, 16, 17, and 18; milk components on d 15, 16, 17, and 18; and plasma

AA on d 18 of each treatment period. Laboratory results were used to calculate weighted average of milk fat and protein concentrations (Nair et al., 2025). Corresponding data from the standardization period were used as a covariate adjustment for the treatment periods. Repeated measures on days were included in the model, excluding BW and AA results (1 observation per treatment-period-cow). An autoregressive covariance structure was chosen based on AIC evaluation. Variables failing to meet the criteria for normal distribution of residuals and homoscedasticity were log-transformed. The analysis of milk protein percent was performed with a log-transformation, with the results reported as arithmetic means of raw data and 95% CI (Yu et al., 2024a,b). The model included

$$Y_{ijkl} = \mu + A_i + P_j + M_k + R_{ijk} + D_l + (M \times D)_{kl} + \beta(\chi)_{ijkl} + e_{ijkl},$$

where Y_{ijkl} = the dependent variable; μ = the overall mean; A_i = the random effect of the i th animal; P_j = the random effect of the j th period; M_k = the fixed effect of the k th RPM treatment (k = CTL1, SM, PT1.0, PT1.25, and PT1.5); R_{ijk} = the replicate effect of ijk th replicate of animal, period, and RPM treatment; D_l = the fixed effect of the l th day as repeated measure; $(M \times D)_{kl}$ = the fixed effect of the k th RPM treatment and l th day interaction; $\beta(\chi)_{ijkl}$ = the covariate effect; and e_{ijkl} = the random error. Differences between means were assessed by pairwise comparison using Tukey's honestly significant difference

Table 4. Experiment 1 feed intake, milk production and composition, and BW for lactating Holstein cows fed different concentrations of rumen-protected Met (RPM; LSM $\pm$ SEM)

Item	Treatment ¹					SEM	P-value
	CTL1	SM	PT1.0	PT1.25	PT1.5		
DMI, kg/d	30.4	29.4	30.0	30.1	29.7	0.99	0.33
Milk, ² kg/d	48.9 ^a	47.7 ^{ab}	46.5 ^b	48.0 ^{ab}	47.4 ^{ab}	1.81	0.03
ECM, ³ kg/d	50.5 ^a	50.3 ^a	48.3 ^b	51.5 ^a	51.1 ^a	2.00	0.04
Lactose, %	4.73 ^a	4.65 ^c	4.70 ^{ab}	4.67 ^{bc}	4.67 ^{bc}	0.03	<0.01
Lactose, ² kg/d	2.30 ^a	2.19 ^{bc}	2.18 ^c	2.26 ^{ab}	2.23 ^{abc}	0.08	0.02
True protein, ² %	3.06 ^c	3.22 ^a	3.10 ^{bc}	3.11 ^{bc}	3.18 ^{ab}	—	0.01
True protein, kg/d	1.46 ^a	1.48 ^a	1.41 ^b	1.47 ^a	1.46 ^a	0.05	0.03
Fat, %	3.74	3.82	3.74	3.90	3.86	0.12	0.46
Fat, kg/d	1.81	1.79	1.73	1.89	1.88	0.09	0.07
SNF, %	8.59 ^b	8.67 ^a	8.60 ^b	8.57 ^b	8.60 ^b	0.05	<0.01
SNF, ² kg/d	4.18	4.11	3.98	4.14	4.10	0.15	0.06
MUN, mg/dL	11.8	12.6	12.5	12.3	11.8	0.54	0.08
SCC, ² $\times 10^3$ cells/mL	319	250	308	208	188	118	0.16
Initial BW, kg	671	664	610	656	730	37.8	0.39
Final BW, kg	708	701	684	714	780	58.9	0.81

^{a-c}Within rows, differing superscripts denote differences ($P \leq 0.05$) between means.

¹CTL1 = control with 0 g of RPM supplemented, SM = 22 g of Met supplemented as RPM (Smartamine M; Adisseo, Alpharetta, GA), PT1.0 = 22 g of Met supplemented as RPM prototype (Balchem Co., Montvale, NJ), PT1.25 = 27 g of Met supplemented as RPM prototype, and PT1.5 = 33 g of Met supplemented as RPM prototype.

²Covariate excluded from analysis ($P > 0.10$). P-value reported is for logarithmic transformation of raw data.

Arithmetic mean (%) associated with 95% CI instead of SEM, CTL2 (4.0–2.1), SM6 (4.2–2.3), SM12 (4.0–2.2), PT6 (4.0–2.2), PT12 (4.1–2.1).

³Energy-corrected milk derived from Tyrrell and Reid (1965).

(HSD) procedure. Results are reported as LSM ($\pm$ SEM). Significant differences were declared at $P \leq 0.05$, and tendencies were declared at $0.05 < P \leq 0.10$. Covariate adjustment was excluded from the model when the covariate effect was not significant ($P > 0.10$). There were no effects of day and treatment by day interactions (data not shown).

Experiment 2

Animals, Experimental Design, and Treatment. The Institutional Animal Care and Use Committee of the University of Tennessee approved the experimental procedures of this trial (2559-0917). Experiment 2 used 20 multiparous Holstein cows (arithmetic mean $\pm$ SD: DIM = 72 ± 43 , and milk yield = 49.2 ± 2.57 kg/d at the beginning of the study) from the East Tennessee AgResearch and Education Center, Little River Animal and Environmental Unit lactating herd. Cows were housed in the main barn equipped with sand-bedded freestalls. Cows were stratified based on DIM, milk yield, and parity. Subsequently, cows were assigned randomly to a quadruplicate 5×5 Latin square. The length of each period was 14 d (10 d of adjustment to treatments and 4 d of data collection). Treatments were as follows: (1) control (CTL2; 0 g/d of RPM supplemented), (2) 6 g/d of metabolizable Met supplemented as RPM (SM6; Smartamine M), (3) 12 g/d of metabolizable Met supplemented as Smartamine M (SM12), (4) 6 g/d of metabolizable Met supplemented

as RPM prototype 2 (PT6), and (5) 12 g/d of metabolizable Met supplemented as RPM prototype 2 (PT12). Prototype 2 contained 72% (DM basis) of D,L-Met and 28% fat (used as the protective coating from Balchem Corp.) with predicted bioavailability of 57% (Balchem Corp., Montvale, NJ). The prediction in bioavailability was based on Met concentration and metabolizable Met in the prototype using a 3-step in vitro procedure (Ross et al., 2013). The RPM products were top-dressed in the TMR twice daily at ~0700 and ~1700 h (half and half of the daily allocation) and mixed into the top third portion of the diet. The formulation of the basal diet targeted a Holstein cow with 669 kg BW, 28.0 kg/d DMI, and 47.6 kg/d milk yield with 4.30% milk fat and 3.06% milk protein (NDS Professional).

The basal diet was formulated to contain 16.0% CP (DM basis) at a 52:48 forage-to-concentrate ratio with a balance of -9.5 g/d metabolizable Met (2.02% of MP; Table 1). The forage portion of the diet was corn and ryegrass silages, and the concentrate portion consisted of corn grain, soybean meal, protected soybean meal (SoyPLUS, Landus, Des Moines, IA), soybean hulls, and citrus pulp. During a 14-d standardization period before the start of treatments, cows were fed the basal diet supplemented with 12 g/d of metabolizable Met using Smartamine M. The basal diet contained 30 g of RP-Lys supplement (Aji-Pro L generation 2, Ajinomoto Co. Inc.) to meet Lys requirements. The basal and treatment diets

were fed to individual cows using an electronic feeding system (American Calan Inc.) once daily at 0700 h, targeting 5% to 10% TMR refusal. Milking was completed daily at 0700 and 1830 h, after which milk yields were collected. At the end of the standardization period and each treatment period, BW and BCS were recorded once for each cow after the 0700 h milking. The BCS was performed by a single individual using a scale of 1 to 5 (Edmundson et al., 1989).

Sample Collection and Analyses. Forage, grain mix, and TMR sample collection and analysis were conducted as in experiment 1, as stated previously. The results are reported in the same tables reported for experiment 1. Milk samples were collected at each milking consecutively on d 11, 12, 13, and 14 of the standardizations and each treatment period. The analysis and calculations for milk samples were performed with similar methodology as in experiment 1.

Blood samples from coccygeal vessels and mammary abdominal vein (10 mL each) were collected at 0700 and 0800 h on d 14 of the standardization period and each treatment period. The analyses of AA concentrations were conducted as described for experiment 1. The intra-assay AA CV was 7.4%. Upon removal of outlier values, the interassay CV for mammary vessel and coccygeal vessel was 27.2% and 33.2%. The large variability could have been associated with the day-to-day variability among technicians running the assays and the overall stability of the kit. Methionine, Phe, and Tyr showed a 2.9%, 2.9%, and 3.2% interassay CV, respectively.

Calculations and Statistical Analyses. For experiment 2, the plasma flow of AA was calculated using the Fick principle and Phe and Tyr as markers for mammary plasma flow estimation, as reported by Mephram (1982) and Wang et al. (2016). Mammary clearance rates were estimated using the mammary plasma flow of Phe and Tyr and the difference of AA in plasma (i.e., coccygeal plasma subtracted from mammary abdominal vein concentrations) reported by Hanigan et al. (1998b) and summarized by Rius et al. (2010).

Similar to experiment 1, experiment 2 used a quadruplicate 5×5 Latin square experimental design balanced for residual error (with repeated measures) to determine the effects of RPM supplementation in dairy cattle. The experimental unit was the individual cow. Production data on d 11, 12, 13, and 14 and plasma AA results on d 14 of each treatment period were used in the analysis. Weighted averages for milk fat and protein concentrations were used (Nair et al., 2025). The effects of supplementing RPM were tested using the Mixed procedure in SAS (version 9.4; SAS Institute Inc., Cary, NC). As in experiment 1, corresponding data from the standardization period were used as a covariate adjustment for the treatment periods. Repeated measures on day were included in the

Table 5. Experiment 2 analyzed nutrient and chemical composition of the feed ingredients used in the basal diet (% of DM)

Item	Ingredient ¹		
	Corn silage	Ryegrass silage	Grain mix
DM, % of feed	31.4	26.8	90.2
NDF	43.1	61.4	17.9
ADF	27.0	45.4	13.5
CP	7.72	8.76	20.8
Lignin	3.22	5.90	1.56
Starch	29.1	2.24	22.3
Crude fat	3.18	2.77	4.17
NEL, Mcal/kg	1.65	1.26	1.76
Ca	0.21	0.56	1.74
P	0.26	0.28	0.49
Mg	0.19	0.22	0.47
K	0.97	2.13	1.67
Na	0.01	0.13	0.97

¹Grain mix contained all ingredients listed in Table 1 in addition to corn silage and ryegrass silage.

model, excluding BW, BCS, and AA results. There were no effects of time or treatment by time interactions detected, so main effects are reported. Post hoc analyses of treatment means were conducted to determine differences using Tukey's HSD procedure. The same statistical model and significance parameters were used as described in experiment 1. The analyses of SCC and mammary clearance of Met were performed using a log-transformation of data, and results were reported as arithmetic means of raw data with 95% CI. The mammary clearance of Ala, Asp, Glu, Ile, Leu, and Val revealed analytical problems and was not reported. Covariate adjustment was excluded from the statistical model when the covariate effect was not significant ($P > 0.10$).

RESULTS AND DISCUSSION

Dietary Nutrient Composition

In experiment 1, the chemical composition and nutrient contents of ingredients are listed in Tables 2 and 3 (AA analysis).

In experiment 1, the basal diet was formulated at 16.7% CP; however, after adjustments of nutrient composition of feed ingredients and dietary treatments, the basal diet (CTL1) provided 16.2% CP (DM basis; data not shown). Furthermore, the CTL1 diet was formulated using 27.0 kg/d DMI, but the observed DMI was 30.4 kg/d ($\Delta = +3.40$ kg/d; Table 4).

In experiment 2, the chemical composition and nutrient contents of ingredients are listed in Table 5. Relative to the formulated CP level, there was a negligible difference in dietary CP content between the formulated 16.0% CP (Table 1) and the observed 15.8% CP (DM basis, data not shown). The treatment diets were balanced using a

Table 6. Experiment 2 intake, milk production and composition, and BW and BCS for lactating Holstein cows fed different concentrations of 2 rumen-protected Met (RPM) supplements (LSM $\pm$ pooled SEM)

Item	Treatment ¹					SEM	P-value
	CTL2	SM6	SM12	PT6	PT12		
DMI, kg/d	25.5	25.6	25.8	25.8	25.1	0.46	0.87
Milk, kg/d	38.6	38.2	37.6	37.4	38.1	0.70	0.73
ECM, ² kg/d	43.6	44.0	44.2	43.0	43.6	0.96	0.92
Lactose, %	4.79	4.80	4.76	4.76	4.78	0.02	0.65
Lactose, kg/d	1.85	1.81	1.78	1.77	1.82	0.04	0.57
True protein, %	3.08 ^b	3.19 ^{ab}	3.28 ^a	3.17 ^{ab}	3.18 ^{ab}	0.05	0.05
True protein, kg/d	1.18	1.19	1.21	1.18	1.21	0.03	0.80
Fat, %	4.40	4.56	4.66	4.49	4.41	0.09	0.17
Fat, kg/d	1.69	1.70	1.74	1.68	1.69	0.05	0.89
SNF, %	8.73	8.85	8.89	8.79	8.82	0.04	0.11
SNF, kg/d	3.36	3.33	3.31	3.27	3.35	0.06	0.84
MUN, mg/dL	11.2	10.6	11.2	10.9	11.0	0.48	0.87
SCC, ³ $\times 10^3$ cells/mL	202 (133–273)	200 (130–270)	239 (159–321)	368 (246–489)	294 (390–196)	—	0.35
BW, kg	693	683	692	685	694	9.12	0.53
BCS	2.83	2.86	2.85	2.90	2.89	0.07	0.36

^{a,b}Within rows, differing superscripts denote differences ($P \leq 0.05$) between means.

¹CTL2 (0 g/d of RPM supplemented), 6.0 g/d of metabolizable Met supplemented as RPM (SM6; Smartamine M [Adisseo]), 12.0 g/d of metabolizable Met supplemented as Smartamine M (SM12), 6.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT6), and 12.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT12).

²Energy-corrected milk calculated in equation derived from Tyrrell and Reid (1965).

³P-value reported is for logarithmic transformation of raw data. Arithmetic means of raw data (cells/mL) associated with 95% CI (in parentheses) instead of SEM.

DMI estimated at 28.0 kg/d, but results showed a DMI of 25.5 kg/d ($\Delta = -2.50$ kg/d).

Milk Production and Productivity

In experiment 1, there was no effect of day and day-by-treatment interaction, so the main effect of treatment is shown. Treatments did not affect DMI. The SM treatment had similar milk yield to CTL1 (47.7 vs. 48.9 kg/d; Table 4). Compared with CTL1, PT1.0 decreased milk yield by 2.40 kg/d ($P = 0.03$). Compared with CTL1, PT1.25 and PT1.5 supported similar milk yields (48.9, 48.0, and 47.4 kg/d). In comparison with CTL1, SM, PT1.25, and PT1.5 had similar ECM yields, whereas PT1.0 decreased ECM yield (48.3 vs. 50.5 kg/d; $P = 0.04$). Compared with CTL1, PT1.0 showed similar lactose concentration, but SM, PT1.25, and PT1.5 had lower lactose concentration ($P < 0.01$). Compared with CTL1, PT1.25 and PT1.5 showed similar lactose yield, but SM and PT1.0 had lower lactose yield ($P = 0.02$). Treatments did not affect the concentration and yield of milk fat. In comparison with the literature, a meta-analysis showed that the supplementation of RPM in dairy cow rations did not affect milk yield and lactose concentrations (Wei et al., 2022). Another study using early-lactation cows fed increasing amounts of RPM from 0 to 72 g/d showed no linear or quadratic effects on milk yield and milk lactose percent and yield (Berthiaume et al., 2006). Regarding other milk components, studies reported that Smartamine M supplementation increased

milk fat percent and yield (Noftsgger and St-Pierre, 2003; Socha et al., 2005); however, treatments did not affect these parameters in the current study. We concluded that CTL1 and RPM treatments had a greater than expected DMI, and it is possible that diets supplied an excess of metabolizable Met, nullifying the effect of additional supply of Met in RPM treatments. Therefore, discrepancies between formulated and observed DMI may have limited the response to RPM supplementation on milk component concentrations and yields.

In experiment 1, treatments affected milk protein percent (Table 4). Overall, the RPM supplementation increased the milk protein percent in the SM treatment by 0.13 percentage units compared with CTL1 (3.13% vs. 3.00%). The PT1.5 treatment was not different from the SM treatment, but it was also not different from PT1.0 and PT1.25 treatments. The PT1.0 and PT1.25 treatments reported similar milk protein content as the CTL1. In comparison with other published studies that supplemented RPM in moderate CP dairy rations, cows were supplemented with 9.9 and 7.7 g/d of Smartamine M and resulted in an increase in milk protein by 0.06 percentage units (3.30% vs. 3.24%) and 0.10 percentage units (3.09% vs. 2.99%), respectively (Noftsgger and St-Pierre, 2003; Zang et al., 2017). Moreover, supplementation with Smartamine M to a 15.6% CP diet increased milk protein percent and yield by 0.12 percentage units (3.15% vs. 3.03%) and 0.09 kg/d (1.33 vs. 1.24 kg/d) compared with no supplementation (Chen et al., 2011). Methionine is a

Table 7. Experiment 1 plasma AA concentrations (μM) in coccygeal vessels of lactating Holstein cows fed varying amounts of rumen-protected Met (RPM; LSM $\pm$ SEM)

Item ²	Treatment ¹					SEM	P-value
	CTL1	SM	PT1.0	PT1.25	PT1.5		
Ala	189	160	178	170	182	11.6	0.36
Arg	90.0	100	96.0	85.9	86.9	6.40	0.34
Asn ³	65.0	60.9	66.0	59.3	60.3	5.78	0.53
Asp ³	8.75	8.65	8.60	9.16	9.35	0.92	0.83
Cit ³	51.7	52.9	47.8	47.0	50.8	6.38	0.55
Glu ³	68.5	56.0	62.9	71.7	72.0	7.16	0.12
Gln ³	187	191	185	174	196	16.2	0.34
His ³	60.0	45.3	55.9	52.8	57.4	4.12	0.08
Ile ³	120	110	120	122	115	9.72	0.78
Leu	117	110	116	116	112	10.8	0.94
Lys ³	138	133	127	127	129	10.6	0.84
Met	27.6 ^b	37.4 ^a	31.5 ^b	31.3 ^b	32.0 ^b	1.96	<0.01
Orn	91.8	81.7	76.8	79.8	81.2	9.86	0.54
Phe ³	45.1	44.6	43.6	42.5	46.0	3.15	0.87
Pro	87.9	80.8	88.6	85.5	85.2	5.34	0.83
Ser	107	91.9	93.0	87.2	96.8	9.50	0.06
Tyr	54.7	51.0	50.8	47.7	49.2	3.53	0.65
Trp	56.3	49.0	49.9	49.0	50.6	3.15	0.25
Val	225	195	214	208	205	17.0	0.60
BCAA	461	417	450	445	433	35.2	0.77
URCY	234	235	220	213	219	14.6	0.55
Total EAA	881	822	851	835	847	50.1	0.88
Total NEAA	942	857	881	859	908	47.0	0.28

^{a,b}Within rows, differing superscripts denote differences ($P \leq 0.05$) between means.

¹CTL1 = control with 0 g of RPM supplemented, SM = 22 g of Met supplemented as RPM (Smartamine M; Adisseo, Alpharetta, GA), PT1.0 = 22 g of Met supplemented as RPM prototype (Balchem Co., Montvale, NJ), PT1.25 = 27 g of Met supplemented as RPM prototype, and PT1.5 = 33 g of Met supplemented as RPM prototype.

²BCAA = Ile + Leu + Val; URCY = Arg + Cit + Orn.

³Covariate excluded from analysis ($P > 0.10$).

key EAA regulating the initiation of protein synthesis. An increased supply of Met reaching mammary secretory cells can increase the synthesis of milk proteins by activating the mammalian target of rapamycin (mTOR) signaling complex (Arriola Apelo et al., 2014; Batistel et al., 2017; Duan et al., 2017). The mTOR signaling cascade promotes the translation of milk proteins in bovine mammary epithelial cells (Wang and Proud, 2006). Thus, RPM may stimulate the activation of mTOR signaling and promote milk protein synthesis. Collectively, the study indicates that RPM supplementation can increase milk protein concentration, and the prototype PT1.5 and SM supported similar milk protein response.

In experiment 2, there was no effect of time or time-by-treatment interaction, so only the main effect of treatment is shown. Supplementing with RPM increased ($P = 0.05$) milk protein content (Table 6). When observing the effect of the Smartamine M treatments compared with CTL2, only SM12 increased ($P < 0.03$) milk protein content by 0.2 percentage units (3.28% vs. 3.08%). The effect of Smartamine M supplementation on milk protein content has been well documented in the literature (Socha et al., 2005; Chen et al., 2011). Concerning the effect of the prototype, compared with CTL2, treatments PT6

and PT12 had similar milk protein content (3.17% and 3.18%, respectively). Compared with SM6 and SM12, treatments PT6 and PT12 showed similar milk protein content. In agreement with the present results, cows supplemented with 15.8 g/d of RPM increased milk protein content by 0.09 percentage units compared with cows not supplemented with RPM (Leonardi et al., 2003). Furthermore, supplementation with 15.0 g/d of RPM as Smartamine M increased milk protein content by 0.21 percentage units compared with a basal diet with no RPM supplementation (Socha et al., 2005). Collectively, RPM supplementation increased milk protein content, which could be associated with an increase in metabolizable Met supply and Met availability to the mammary glands. In summary, SM12 increased milk protein concentration compared with CTL2, and the prototype treatments provided metabolizable Met that sustained milk protein content similar to SM6 and SM12 treatments.

Plasma Amino Acid Concentrations and Amino Acid Mammary Clearance and Uptake

In experiment 1, plasma Met concentrations in coccygeal vessels were not affected by the prototype treat-

Table 8. Experiment 2 plasma AA concentrations (μM) in coccygeal vessels of lactating cows fed different amounts of 2 rumen-protected Met (RPM) supplements (LSM $\pm$ pooled SEM)

Item ²	Treatment ¹					SEM	P-value
	CTL2	SM6	SM12	PT6	PT12		
Ala	325 ^b	327 ^{ab}	388 ^{ab}	397 ^a	397 ^a	36.6	0.05
Arg	103	97.8	108	107	112	9.37	0.18
Asn	33.0 ^a	21.0 ^c	23.5 ^{bc}	26.8 ^b	25.7 ^{bc}	4.25	<0.01
Asp	8.18 ^b	10.8 ^a	11.8 ^a	13.1 ^a	9.24 ^{bc}	1.62	<0.01
Cit	85.5 ^a	67.1 ^b	74.4 ^b	71.5 ^b	73.0 ^b	9.21	<0.01
Gln	132 ^a	135 ^a	113 ^{bc}	124 ^{ab}	107 ^c	12.7	<0.01
Glu	71.7	77.5	73.3	80.5	78.6	6.34	0.63
His	40.0	43.4	47.3	41.2	37.4	5.83	0.29
Ile	135 ^b	130 ^b	127 ^b	178 ^a	131 ^b	14.1	<0.01
Leu	151 ^b	149 ^b	150 ^b	201 ^a	159 ^b	16.6	<0.01
Lys	107 ^a	95.6 ^{ab}	99.8 ^{ab}	95.7 ^{ab}	85.2 ^b	7.22	0.02
Met	22.0 ^b	23.8 ^b	27.7 ^a	23.5 ^b	24.3 ^b	1.27	<0.01
Orn	65.6 ^a	52.3 ^{bc}	59.3 ^{ab}	54.2 ^{bc}	51.2 ^c	4.68	<0.01
Phe	50.7	45.6	49.8	50.9	50.6	4.17	0.15
Pro	121	119	129	142	123	9.88	0.19
Ser	80.5	71.8	78.4	69.8	75.2	4.89	0.21
Trp	39.1	37.4	36.3	37.8	34.6	3.54	0.38
Tyr	49.2	43.7	46.0	45.1	46.3	2.97	0.31
Val	287 ^c	300 ^c	386 ^{ab}	425 ^a	347 ^{bc}	35.7	<0.01
BCAA	557 ^{cd}	481 ^d	664 ^b	786 ^a	580 ^{bc}	58.5	<0.01
URCY	254 ^a	217 ^c	242 ^{ab}	233 ^b	232 ^b	13.6	0.01

^{a-d}Within rows, differing superscripts denote differences ($P \leq 0.05$) between means.

¹CTL2 (0 g/d of RPM supplemented), 6.0 g/d of metabolizable Met supplemented as RPM (SM6; Smartamine M [Adisseo]), 12.0 g/d of metabolizable Met supplemented as Smartamine M (SM12), 6.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT6), and 12.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT12).

²BCAA = Ile + Leu + Val; URCY = Arg + Cit + Orn.

ments. The SM treatment increased plasma concentrations of Met in coccygeal vessels compared with the CTL1, PT1.0, PT1.25, and PT1.5 treatments by 10, 6, 6, and 5 μM , respectively ($P \leq 0.01$; Table 7). Compared with our study, previous work using a Met-deficient diet showed that supplementing RPM to increase metabolizable Met by 22 and 43 g/d and to exceed the established methionine requirements increased plasma Met concentrations by 4 and 15 μM , respectively (Berthiaume et al., 2006). In summary, supplementation of RPM with the SM treatment increased plasma concentrations of Met, and this response was paralleled with a positive response in milk protein content.

In experiment 2, RPM supplementation with SM and prototype treatments increased ($P \leq 0.01$) Met concentration in plasma (Tables 8 and 9). Specifically in plasma harvested from coccygeal vessels, the SM12 treatment increased ($P \leq 0.01$) Met concentration by 26% compared with CTL2 (27.7 vs. 22.0 μM) and by up to 18% compared with the SM6, PT6, and PT12 treatments. The PT12 treatment numerically increased Met concentration by 10.5% compared with CTL2 (24.3 vs. 22.0 μM). The concentration of metabolites in plasma harvested from coccygeal vessels is used to measure metabolite concentrations in arterial plasma and perform arterio-venous difference studies in dairy cattle. Regarding plasma obtained

from the mammary abdominal vein, the SM6, SM12, and PT12 treatments increased ($P < 0.01$) Met concentration by up to 74% (Table 9) when compared with the CTL2 treatment. Plasma Met concentrations of CTL2 and PT6 treatments did not differ. Treatment SM6 increased ($P < 0.01$) Met concentration by 35% compared with PT6 (13.4 vs. 9.93 μM), whereas SM12 increased ($P < 0.01$) Met concentration by 24% compared with PT12 (14.9 vs. 12.0 μM). Treatment PT12 tended to increase ($P = 0.06$) plasma Met concentrations compared with PT6. A study comparing RPM products showed that SM increased serum Met (27.3 μM) more effectively than Mepron 85 (23.3 μM ; Evonik Industries, Essen, Germany) or Novimet (22.7 μM ; Innovad, Essen, Belgium) in comparison with a control diet (21.2 μM ; Zang et al., 2017). Serum Met concentrations and the change in serum Met concentrations to RPM supplementation, as shown by Zang et al. (2017), agree with our study. Intriguingly, the literature shows a positive response, such that a 50 g/d Met dietary supplementation using Smartamine M increased plasma Met concentrations by 128 μM (Blum et al., 1999); however, this concentration is ~4-fold higher than the typical plasma concentrations observed following supplementation with rumen-protected methionine. Collectively, these results suggest that the supplementation of Met using Smartamine M and prototype 2 at 12 g/d of metaboliz-

Table 9. Experiment 2 plasma AA concentrations (μM) in the abdominal mammary vein of lactating Holstein cows fed different amounts of 2 rumen-protected Met (RPM) supplements (LSM $\pm$ pooled SEM)

Item ²	Treatment ¹					SEM	P-value
	CTL2	SM6	SM12	PT6	PT12		
Ala	359	312	347	309	308	64.8	0.47
Arg	78.1 ^{ab}	75.6 ^{ab}	72.7 ^b	68.8 ^b	85.7 ^a	6.10	0.02
Asn	28.9	33.6	31.6	31.2	30.8	5.45	0.59
Asp	9.62 ^b	9.31 ^b	7.86 ^c	11.8 ^a	10.8 ^a	2.67	<0.01
Cit	59.0 ^{bc}	53.1 ^c	73.3 ^a	59.5 ^{bc}	47.3 ^{ab}	5.46	0.05
Gln	150 ^b	157 ^b	160 ^{ab}	143 ^b	179 ^a	18.6	<0.01
Glu	49.8	54.5	49.9	61.5	51.6	10.5	0.14
His	44.2	41.7	48.6	42.4	41.9	4.28	0.60
Ile	85.1	117	123	107	128	20.7	0.11
Leu	89.5 ^b	115 ^{ab}	122 ^a	133 ^a	121 ^{ab}	16.8	0.02
Lys	70.0	68.1	58.3	63.7	71.0	7.33	0.19
Met	8.54 ^d	13.4 ^{ab}	14.9 ^a	9.93 ^{cd}	12.0 ^{bc}	1.13	<0.01
Orn	46.7	47.0	38.7	42.1	48.8	6.24	0.06
Phe	25.0	25.1	26.8	24.2	27.6	2.50	0.51
Pro	111	90.1	112	99.3	90.1	18.7	0.07
Ser	88.1 ^a	69.9 ^b	68.2 ^{bc}	61.3 ^c	73.8 ^b	4.16	<0.01
Trp	55.2 ^{bc}	55.8 ^b	51.1 ^{bc}	50.7 ^c	62.6 ^a	4.24	<0.01
Tyr	31.9	31.0	27.8	27.2	33.2	3.00	0.12
Val	302	273	281	282	234	56.8	0.16
BCAA	544	548	492	594	551	75.9	0.63
URCY	183 ^b	176 ^b	187 ^{ab}	172 ^b	206 ^a	13.2	0.03

^{a-d}Within rows, differing superscripts denote differences ($P \leq 0.05$) between means.

¹CTL2 (0 g/d of RPM supplemented), 6.0 g/d of metabolizable Met supplemented as RPM (SM6; Smartamine M [Adisseo]), 12.0 g/d of metabolizable Met supplemented as Smartamine M (SM12), 6.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT6), and 12.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT12).

²BCAA = Ile + Leu + Val; URCY = Arg + Cit + Orn.

able Met was paralleled with increased Met in systemic circulation and available to the mammary glands.

In experiment 2, plasma flow (Table 10) and mammary uptake of AA (Table 11) were not affected by treatments. It is possible that using a single sampling point on the last day of each period and the timing relative to ration allocation reduced the accuracy in measuring plasma flow. For example, the literature shows that multiple and continuous plasma measurements of AA immediately after feeding and determined on 2 to 3 continuous days of the experiment resulted in reasonable variability (Rius et al., 2010; Whitehouse et al., 2016). Additionally, in our study, the commercial kit used to determine AA concentrations showed elevated variability in some AA. This single issue could have reduced accuracy in our plasma flow and mammary clearance measurements. In fact, some AA showed elevated variability within determination. Nonetheless, Met concentrations were within a small range of variation. Treatments affected the clearance rate of Met in the mammary glands ($P < 0.01$; Table 10). Compared with CTL2, SM12 reduced the clearance rate of Met in the mammary glands ($P = 0.01$). Compared with SM12, treatments SM6, PT6, and PT12 had a similar clearance rate of Met in the mammary glands. Previous research showed that the mammary glands reduced plasma flow, the clearance rate of EAA, or both when AA are supplied

above requirements to support protein synthesis (Hanigan et al., 1998a; Rius et al., 2010). Compared with the literature, in our study, changes in AA supply should be negligible as DMI did not change and only Met supply was modified. As expected, CTL2 did not increase plasma flow in the udder. However, the mammary glands lowered the clearance rate of Met in response to supplementation of Smartamine M at 12 g. This result highlights the udder responsiveness to RPM supplementation and the changes in Met supply. Collectively, these results suggest that RPM supplementation increased the availability of Met to peripheral tissue, and the mammary glands responded by lowering Met clearance rate.

Concentrations of Arg, Cit, and Orn and their combination (URCY) as contributors to AA catabolism and urea synthesis can be indicative of how the RPM treatments influenced the efficient use of AA. However, use efficiency of AA was not measured in this study, and other factors could have influenced this measurement. Treatments SM6, PT6, and PT12 decreased ($P < 0.01$) plasma concentrations of Cit, Orn, and URCY in the coccygeal vessels compared with CTL2 (Table 8). These results suggest that RPM supplementation improved utilization of AA for anabolism, which may have contributed to greater synthesis of proteins in mammary tissues. In support of this logic, a supplementation with 60 g/d of RPM

Table 10. Experiment 2 mammary plasma flow (L/h) and clearance rate¹ (L/h) of AA from lactating Holstein cows fed different concentrations of 2 rumen-protected Met (RPM) supplements (LSM ± pooled SEM)

Item	Treatment ²					SEM	P-value
	CTL2	SM6	SM12	PT6	PT12		
Plasma flow	722	690	932	743	882	270	0.95
Clearance rate							
Arg	485	609	635	491	342	175	0.69
Asn	338	-442	-324	-34.8	-208	244	0.06
Cit	433	544	307	420	499	299	0.98
Gln	257	-269	-322	96.7	-378	244	0.11
His	700	84.5	-388	87.4	-299	377	0.31
Lys	752	631	770	484	756	227	0.83
Met ³	1,715 ^a	1,389 ^{ab}	886 ^c	1,410 ^{ab}	1,102 ^{bc}	—	0.01
Orn	629	256	692	280	587	202	0.36
Phe	1,021	980	1,009	1,097	1,121	176	0.71
Pro	1,427	409	1,102	896	1,937	758	0.69
Ser	10.9	198	-276	368	-138	154	0.10
Trp	-49.7	-214	-324	-58.9	-367	174	0.31
Tyr	591	730	889	758	739	173	0.83

^{a-c}Within rows, differing superscripts denote differences ($P \leq 0.05$) and trends ($0.05 < P \leq 0.10$) between means.

¹Clearance rate (k_i) = $MPF \times AV_i/[V_i]$ (from Hanigan et al., 1998b and Rius et al., 2010), where k_i is the clearance rate, MPF is mammary plasma flow of Phe + Tyr (L/h), AV_i is the difference of the i th AA from coccygeal vessel minus mammary vein (mol/L), and $[V_i]$ is the plasma concentration of the i th AA from the mammary vein (mol/L). Positive values indicate AA uptake, and negatives values indicate AA output by mammary glands.

²CTL2 (0 g/d of RPM supplemented), 6.0 g/d of metabolizable Met supplemented as RPM (SM6; Smartamine M [Adisseo]), 12.0 g/d of metabolizable Met supplemented as Smartamine M (SM12), 6.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT6), and 12.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT12).

³P-value reported is for logarithmic transformation of raw data. Arithmetic mean (L/h) associated with 95% CI instead of SEM, CTL2 (2,302–1,160), SM6 (1,890–952), SM12 (1,035–521), PT6 (1,826–920), PT12 (1,390–700).

(39 g/d of D,L-Met) lowered plasma urea concentrations, and the portal Met flux had a negative correlation ($r = -0.59$) with plasma urea concentrations (Bach et al., 2000). Therefore, supplying RPM likely increased pro-

tein anabolism while inhibiting AA deamination (Bach et al., 2000). Collectively, these data suggest that supplementing RPM may improve AA metabolism by reducing AA deamination and urea synthesis.

Table 11. Experiment 2 mammary gland AA uptake¹ (mmol/h) in lactating Holstein cows fed different concentrations of 2 rumen-protected Met (RPM) supplements (LSM ± pooled SEM)

Item	Treatment ²					SEM	P-value
	CTL2	SM6	SM12	PT6	PT12		
Arg	17	21	28	33	23	9.1	0.64
Asn	4.2	-19	-17	-4.7	-3.9	8.2	0.09
Cit	17	21	-10	21	26	12	0.28
Gln	36	-21	-66	-26	-53	43	0.35
His	9.4	5.4	-24	-1.0	-22	11	0.24
Lys	27	23	35	21	32	9.0	0.83
Met	10	12	10	10	9.6	1.3	0.55
Orn	14	6.2	16	5.8	15	6.2	0.57
Phe	20	23	25	22	19	2.6	0.52
Pro	60	85	58	61	81	35	0.90
Ser	1.4	10	-15	12	-14	10	0.13
Trp	-2.3	-17	-15	-7.9	-23	10	0.50
Tyr	14	12	11	12	15	2.3	0.77

¹AA uptake = $MPF \times AV_i$ (from Hanigan et al., 1998b and Rius et al., 2010), where MPF is mammary plasma flow of Phe + Tyr (L/h) and AV_i is the difference of the i th AA from coccygeal vessel minus mammary vein (mmol/L). Positive values indicate AA uptake, and negatives values indicate AA output by mammary glands.

²CTL2 (0 g/d of RPM supplemented), 6.0 g/d of metabolizable Met supplemented as RPM (SM6; Smartamine M [Adisseo]), 12.0 g/d of metabolizable Met supplemented as Smartamine M (SM12), 6.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT6), and 12.0 g/d of metabolizable Met supplemented as RPM prototype 2 (PT12).

In experiment 2, the concentration of EAA in coccygeal vessels was affected by treatments such that PT6 increased concentrations of Leu, Ile, Val, and Ile + Leu + Val (BCAA), compared with CTL2. This response was unexpected, and a clear explanation may not be available.

CONCLUSIONS

In experiment 1, supplementing SM and prototype 1 at PT1.5 increased milk protein concentration, and SM increased plasma concentrations of Met. In experiment 2, prototype 2 at both amounts showed similar milk protein concentrations relative to SM6 and SM12. Compared with CTL2, treatment SM12 increased milk protein concentration, and this response was paralleled with greater Met concentration in systemic circulation. Rumen-protected products at both amounts increased the availability of Met to peripheral tissues, which may have contributed to supporting protein synthesis in the udder. Treatment SM12 reduced the clearance of Met in the mammary glands, suggesting the use of additional Met in other tissues. Overall, commercially available RPM and the prototypes elicited an increase in plasma Met concentrations, and this response was associated with an increase in milk protein content.

NOTES

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Nonstandard abbreviations used: BCAA = Ile + Leu + Val; CTL1 = control diet experiment 1; CTL2 = control diet experiment 2; HSD = honestly significant difference; PT = prototype; RPM = rumen-protected Met; SM = Smartamine M; URCY = Arg + Cit + Orn.

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